

B2B Event Ticketing Platform

1. Introduction

This project demonstrates how data from different sources can be collected, stored, transformed, and reported. The goal is to provide business users with reliable dashboards without manual work.

2. Business Problem

Ticket sales come from an operational PostgreSQL database and reseller CSV files. These sources are separate, making reporting difficult. The solution combines them into a single Snowflake data warehouse.

3. Solution Overview

Airflow schedules and monitors all data pipelines. Raw data is loaded into Snowflake, transformed with dbt into clean business tables, and visualized in Power BI.

4. End-to-End Architecture

Neon PostgreSQL → Airflow → Snowflake RAW

GitHub CSV → Airflow → Snowflake RAW

Airflow (dbt build) → Staging → Dimensions → Facts → Power BI

5. Technology Stack

Python for ETL scripts

Neon PostgreSQL as source database

GitHub for reseller CSV files

Apache Airflow for orchestration

Snowflake as cloud warehouse

dbt for transformations

Power BI for reporting

6. Pipeline 1

The `neon_to_snowflake_daily` DAG extracts only new or changed records from PostgreSQL using a watermark. Data is validated before loading into RAW tables.

7. Pipeline 2

The `reseller_csv_github_to_snowflake` DAG checks GitHub for new CSV files, downloads them, validates file structure, prevents duplicate loads using file history, and merges data into Snowflake.

8. Pipeline 3

The `dbt_build_b2b_event_ticketing` DAG executes dbt models. Staging models clean the data, dimension models organize business entities, and fact tables store measurable business events.

9. Data Warehouse Layers

RAW stores source data unchanged.

STAGING standardizes columns and data types.

DIMENSIONS describe customers, events, organizers, resellers, and dates.

FACTS contain ticket sales and revenue.

10. Star Schema

The FACT_TICKET_SALES table links to dimensions including Customer, Event, Organizer, Reseller, and Date. This model provides fast reporting and simple queries.

11. Data Quality

Checks include duplicate detection, null validation, relationship tests, accepted values, incremental loading, and dbt tests.

12. Monitoring

Airflow records execution time, status, row counts, retries, and errors. This makes troubleshooting easier.

13. Power BI Dashboard

Page 1 contains Executive Summary with KPIs, sales trends, and channel analysis. Page 2 provides detailed analysis by organizer, reseller, region, ticket type, and commission.

14. Benefits

Fully automated daily pipeline.
Scalable cloud architecture.
Improved data quality.
Centralized reporting.
Easy maintenance through dbt and Airflow.

15. Future Improvements

Add CI/CD, automated deployments, Snowflake Tasks, Great Expectations, Terraform, alerts, and monitoring dashboards.

16. Conclusion

This project shows a complete modern ELT architecture suitable for analytics projects. The design is modular, maintainable, and easy to extend.